

Supplementary Materials

Transfer learning with simulated variants and calculated quantities for nanobody melting-temperature prediction

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Supplementary methods

Additional details for the diverse MD panel

The diverse panel was assembled from an archived SAbDab summary and the corresponding IMGT-renumbered structures [1]. For each PDB identifier, the workflow took the first heavy-chain entry whose chain identifier was present in the structure file. After availability checks, this deterministic selection yielded 1143 structures with a usable sequence, topology, trajectory, and native-contact label.

The selected heavy chain was kept as an uncapped protein at pH 7.4, solvated in OPC water with 15 Å padding and 0.15 M NaCl, and described with ff19SB [2,3]. The 300-K branch comprised 1 ns NVT and 1 ns NPT with a 2-fs timestep. Two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps⁻¹ friction [4].

The diverse-panel contact calculation used backbone heavy-atom pairs separated by more than three residues, within 0.45 nm in the reference structure, and with at least one atom in Asp, Glu, Gln, Asn, Arg, Lys, or His. For trajectories with more than 300 frames, the label was averaged over the larger of the final 300 frames and the final one third of the trajectory; all frames were used for shorter trajectories. Supplementary Fig. S1 shows the relation between sequence length and the resulting raw Q value.

FEP provenance and charge-change check

The retained FEP tables contain Ala, Asp, Ile, and Gln scans. Row-level provenance was checked against the archived scan outputs; the available Phe scans were not included.

For charge-changing mutations in a cubic 70-Å box and solvent dielectric 78, we tested an analytical periodic net-charge correction of $0.086(\Delta q)^2$ kcal mol⁻¹. We applied the correction before repeating the sign reversal, robust scaling, Yeo–Johnson transform, and min–max scaling used for the model-facing labels (Supplementary Fig. S2b).

References

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- [2] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguette, L., Huang, H., Miguels, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. *Journal of Chemical Theory and Computation* 16, 528–552 (2020). <https://doi.org/10.1021/acs.jctc.9b00591>
- [3] Izadi, S., Anandakrishnan, R., Onufriev, A. V. Building water models: a different approach. *The Journal of Physical Chemistry Letters* 5, 3863–3871 (2014). <https://doi.org/10.1021/jz501780a>
- [4] Eastman, P., Swails, J., Chodera, J. D., McGibbon, R. T., Zhao, Y., Beauchamp, K. A., et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology* 13, e1005659 (2017). <https://doi.org/10.1371/journal.pcbi.1005659>

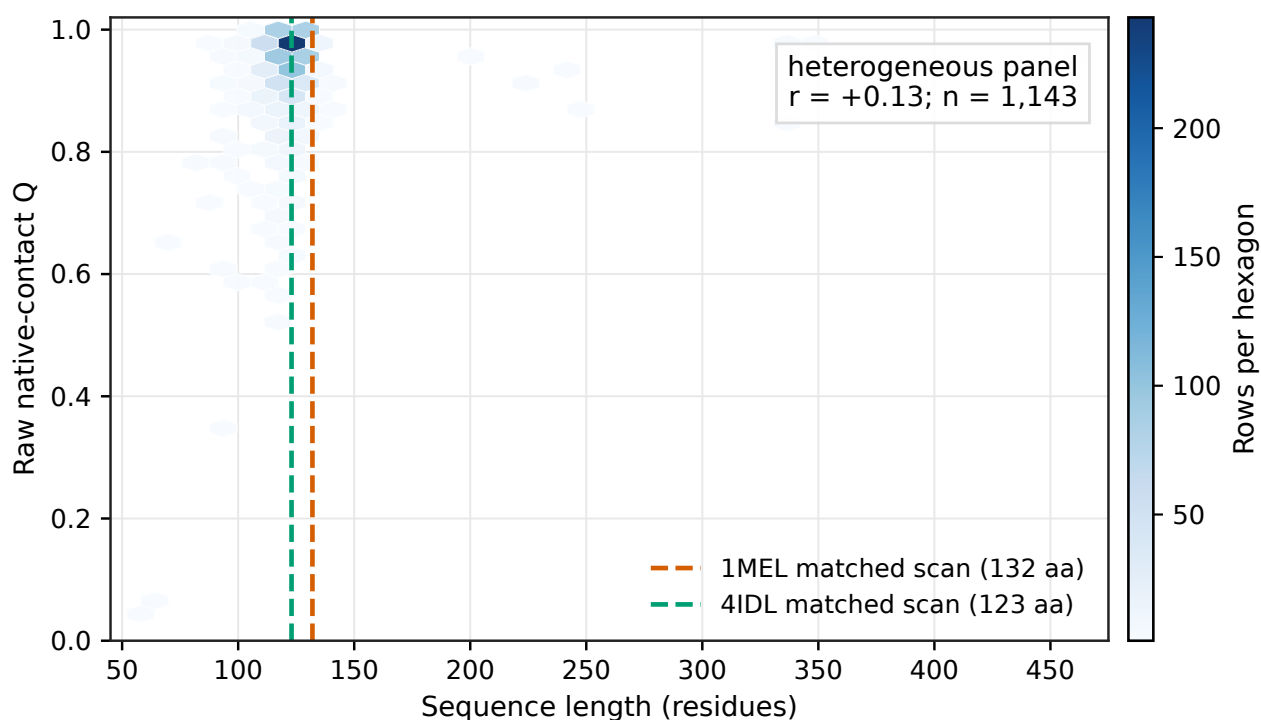


Figure S1 Sequence length and native-contact Q in the heterogeneous MD panel. Hexagons show row density, and dashed lines mark the fixed sequence lengths of the 1MEL and 4IDL matched mutation scans. The observed association is descriptive and does not establish that length caused the transfer difference.

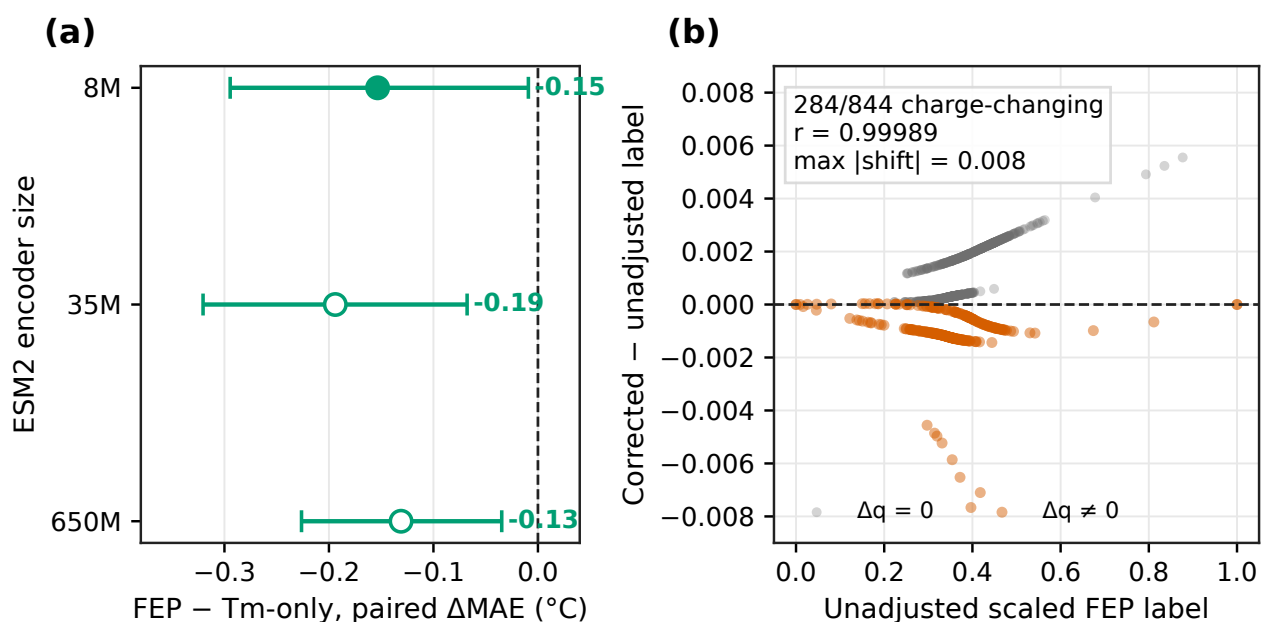


Figure S2 Robustness controls for the FEP result. **(a)** Paired FEP-minus-Tm-only MAE for fine-tuned ESM2 encoders. Negative values mean lower Tm error with FEP. The 8M point is the staged-selected five-seed model; the 35M and 650M points are exploratory fixed-configuration ten-seed controls and were not subjected to the same search. Whiskers are nominal 90% paired bootstrap intervals over the 396 test proteins. **(b)** Change in each scaled FEP label after the analytical periodic net-charge correction at dielectric 78.